ResQ

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Chapter 1

Introduction

Our search and rescue robots use affordance principles, which deal with the robot's interaction with objects, together with path planning. Although it can be dangerous to send human rescuers, saving humans from debris which is an important goal. Our objective is to incorporate idea into multi-agent robots to enhance life-saving operations for humans.

1.1 Problem Domain

Our area of interest focuses on the integration of affordance concepts in multi-agent robotic systems for search and rescue operations,particularly during natural disasters. Affordance is principles that refers to robot's ability to perceive and interact with objects , determining how object can be moved or used to achieve a specific task.

Currently, our research robots indicates a significant gap in the use of the affordance-based approaches in search and rescue robotics. It heavily rely on the predefined path planning ,with limited capacity to adapt to unpredictable or unknown environments. These robot rely heavily on the static maps. All the approaches use in the papers works well in controlled,static setting, it find difficult to work in dynamic environments that are constantly changing, such as those found in disaster zones e.g earthquake.

Moreover, multi-agent robotic systems, are not widely used in field of search and rescue. In typical approach found in existing research, robot often rely on predefined maps that are loaded into their systems before deployment. While it fails to account for unexpected obstacles, damage. These robot are limited in their ability to perform under real-world conditions where flexibility and adaptability are crucial.

Our research seeks to address the gap by developing affordance concepts with multi-agent systems, specifically for unpredictable environment such as earthquake.

1.2 Research Problem Statement

In the aftermath of natural disasters like earthquakes, collapsed buildings create unpredictable and highly congested environments, making rescue operations extremely challenging. Traditional methods, which often rely on single robots, struggle to navigate these complex and dynamic conditions. The unpredictability of the debris and obstacles further complicates the process, slowing down efforts to locate and reach survivors in time. The complexity of such environments poses a significant challenge to current rescue technologies, limiting their effectiveness in critical situations where every second counts.

1.3 Software Requirements Specification (if applicable)

The functional and non-functional requirements for the software that will be used to develop and simulate multi-agent robots using affordance and path planning concepts in Simulator like Gazebo.

1.3.1 Functional Requirements:

1. **Multi-Agent Coordination:**
   - It supports the simultaneous operation of multiple robots within the simulation environment.
   - Robots must be able to communicate and share their information about the environment with each other.
2. **Affordance Concepts:**
   - Robot will be integrated with sensors that allow them to detect affordances of the object.
3. **Path Planning:**
   - After detecting the object, it will update the path in global map , so rescuer can save human easily.

1.3.2 Non-Functional Requirements:

1. **Performance:**

- It must run efficiently,process in real-time interactions.

2. **Scalability:**

   - It must handles more complex environment.

3. **Accuracy**

   - The simulation must provide accurate representations of real-world objects.

Chapter 2

Literature Review

In this chapter, we are going to briefly discuss about the research paper , that we have read so far to identify the research gap and idea to make a solution for over problem.

2.1 Related Research

Here are the analysis of our research papers.

2.1.1 Cerberus Transformer [1]

2.1.1.1 Summary of the Cerberus Transformer

The Cerberus Transformer paper address the challenge of multi-task indoor to understand the scene.It focus on joint semantic, affordance and attribute parsing.The robot handles all the three simultaneously. This model captures long range dependencies and learns from weakly aligned data.It makes use of a customized training architecture that maintains sub task balance and even works effectively with minimal supervision, using only 0.1% of the annotations. Additionally, the study emphasizes the idea of task affinity, which is consistent with human cognition and enhances performance.

2.1.1.2 Critical analysis of the Cerberus Transformer

This paper stands out for its uniqueness and multitasking strategy, which provides a full solution to indoor scene interpretation. The model can effectively handle long-range dependencies thanks to the usage of transformers, which is a powerful decision. Working with badly aligned data, a common problem in multi-task learning, is one of its significant strengths. One drawback, though, is that the model's effectiveness is heavily reliant on the task weights being balanced correctly, which necessitates meticulous optimization. Al-

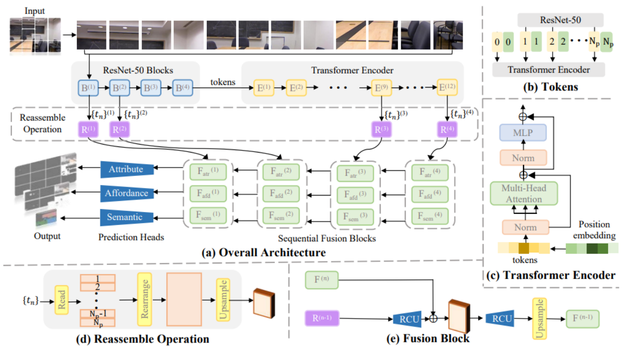

Figure 2. **Overall network architecture of Cerberus.** Given an image, ResNet-50 extract features from the input image to form a set of tokens. The tokens are processed by a transformer encoder and decoded by reassemble operations and fusion blocks. Through three prediction heads, the feature maps are turned into final attribute, affordance and semantic parsing results.

Figure 2.1: Cerberus Transformer

though the results are striking, particularly under limited supervision, the training process of the methodology could demand a large amount of processing power.

2.1.1.3 Relationship to the proposed research work

The Cerberus approach to multi-task learning, that focuses on joint parsing of attributes,affordances and semantics, is highly related to our project. It demonstrates the potential for improving scenes understanding, which is very useful in search and rescue missions where robots will interact with various objects and obstacles As it uses the GenAI concepts transformer. Transformer enhances the ability of robot to process complex information. Cerberus success in handling weak supervision aligns with the challenges that are faced during disaster. It's concept of task affinity, that is how the overall performance of a system, can be increased, will be helpful in for providing coordination of our multi-agents robots.

2.1.2 One-shot Open Affordance Learning with Foundation Models[2]

2.1.2.1 Summary of the research item

This paper introduces the concept of One-shot Open Affordance Learning (OOAL), where model learns addordances using one example per base object category. Its goal was to recognize affordances and generalize to one base objects with minimal training data.The authors evaluate foundation models (such as CLIP and DINOv2) for their ability to learn affordances and suggest a vision-language framework for aligning visual and textual embedding for affordance learning.

2.1.2.2 Critical analysis of the research item

The most important point of this papers lies in its efficient use of limited data and its ability to generalize a novel affordances.The work proposes a novel approach to affordance

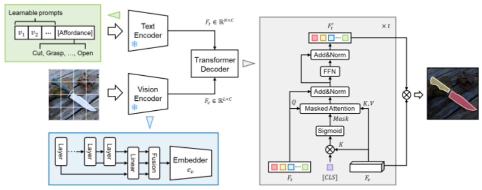

Figure 2.2: OOAL

learning without large-scale datasets by utilizing foundation models and text prompts. However, one disadvantage is that, while the model thrives with little data, it may suffer in cases where more complex or unknown affordances necessitate deeper reasoning than part-based learning. Furthermore, relying on foundation models such as CLIP can induce bias in affordance comprehension due to intrinsic limits in these models' knowledge of object-part interactions.

2.1.2.3 Relationship to the proposed research work

The OOAL technique is quite relevant to our research, particularly in terms of affordance identification in multi-agent robotic systems. The paper's technique, particularly the integration of vision and language models, sheds light on how affordances might be learned with limited data. This is especially crucial in search and rescue operations, since robots frequently have to operate in environments with limited prior knowledge or annotations. The OOAL study gives useful information for improving our affordance detection method. Its usage of foundation models and text-based cues can help improve transfer learning across object types and environments. Furthermore, the vision-language integration is consistent with our goal of using multi modal inputs to improve decision-making. This method can help our multi-agent robotic systems overcome data scarcity difficulties while also improving affordance recognition and generalization.

2.1.3 AffordanceNet: An End-to-End Deep learning Approach for Object Affordance Detection.[3]

2.1.3.1 Summary of the research item

This paper present the concept of AffordanceNet that is deep learning framework designed for real-time detection of both objects and their affordance from RGB images. I t has two branches one is object detection and other is affordance detection. AffordanceNet uses a deconvolution layers and a multi-task loss function.It identify the multiple affordance of object,achieve high accurarcy.

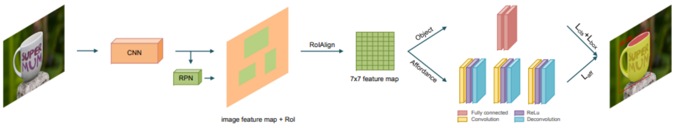

Fig. 2. An overview of our AffordanceNet framework. **From left to right:** A deep CNN backbone (i.e., VGG) is used to extract image features. The RPN shares weights with the backbone and outputs RoIs. For each RoI, the RoIAlign layer extracts and pools its features (from the image feature map, i.e., the $conv5\_3$ layer of VGG) to a fixed size 7×7 feature map. The object detection branch uses two fully connected layers for regressing object location and classifying object category. The object affordance detection branch consists of a sequence of convolutional-deconvolutional layers and ends with a softmax layer to output a multiclass affordance mask.

Figure 2.3: AffordanceNet

2.1.3.2 Critical analysis of the research item

AffordanceNet's primary characteristics are its efficiency and precision in affordance identification, making it ideal for real-time applications, particularly in robotics. Unlike previous approaches that used sequential models, it minimizes the complexity of training and testing by merging object and affordance identification in an end-to-end way. However, one disadvantage is its reliance on specified affordance classes, which may limit its capacity to generalize in more dynamic or unpredictable settings. Furthermore, although the model provides real-time performance, it may need large processing resources, making adoption on less powerful hardware difficult.

2.1.3.3 Relationship to the proposed research work

AffordanceNet's real-time object affordance detection capability is extremely pertinent to our study on multi-agent robots that use affordance identification for search and rescue tasks. AffordanceNet's design provides insights into how affordance detection might be enhanced for quicker decision-making and interaction with items like debris, doors, or other barriers in situations when speed and accuracy are crucial, such as dynamic catastrophe scenarios. Our aim of combining affordance recognition with effective path planning is in line with the multi-task loss function of the model. In order to fully comprehend a scenario and carry out search and rescue operations, AffordanceNet's affordance segmentation method is needed. Its methods for managing pixel-by-pixel fine-grained object affordances give high-stakes tasks the necessary accuracy. In summary, AffordanceNet offers a solid basis for enhancing our multi-agent robotic system's affordance identification skills in real time, guaranteeing accuracy and speed in life-saving tasks in intricate, ever-changing settings.

2.1.4 APEX: Affordance-Based Plan Executor for Indoor Robotic Navigation[4]

2.1.4.1 Summary of the research item

This paper introduces the Affordance-Based Plan Executor (APEX), a novel method for executing navigation plans in indoor environments without the need for a pre-built map. APEX leverages the concept of navigation affordances to switch between high-level behaviors, such as 'turn left' or 'follow a corridor,' as the robot navigates. Using deep learning, APEX identifies the affordances available to the robot through real-time depth images and selects the next step in the plan based on these affordances. Experiments with the Gibson simulator and the Stanford 2D-3D-S dataset demonstrate APEX's ability to outperform alternative methods by up to 43% in success rates, especially in previously seen environments

2.1.4.2 Critical analysis of the research item

A strength of APEX lies in its affordance-based navigation, enabling robots to make decisions based on environmental possibilities, thus bypassing the need for a detailed map. This makes the system highly adaptable, especially in dynamic or unfamiliar environments. However, one limitation is that the system shows decreased performance in unseen environments, particularly when navigation affordances are difficult to identify due to structural differences from the training data. Another challenge is the learning of high-level behaviors, which are essential to APEX's success but may introduce errors if the behavior networks are not robustly trained.

2.1.4.3 Relationship to the proposed research work

APEX's affordance-driven navigation system is particularly relevant to research on autonomous robot navigation and real-time decision-making in dynamic environments. Its focus on affordances as a mechanism for plan execution aligns well with efforts to improve adaptive navigation systems that do not rely on pre-existing maps. The deep learning approach used in APEX to recognize environmental features and execute behaviors can offer valuable insights for integrating affordance recognition and behavior-driven navigation in projects where flexibility in navigation is crucial.

2.1.5 Navigation Among Movable Obstacles with Object Localization using Photo realistic Simulation. [5]

2.1.5.1 Summary of the research item

In order to tackle the problem of Navigation Among Movable Obstacles (NAMO), this research suggests a novel system that uses visual input for the detection, localization, and interaction of movable obstacles. The robot is able to avoid or navigate movable obstacles in a warehouse environment because to the system's integration of a photorealistic simulator (NVIDIA Isaac Sim) for training and real-world robot implementation. The authors utilize obstacle pushing operations, real-time obstacle movability recognition, and graph-based path planning to create free paths. A wheeled robot was used for both simulation and real-world testing.

2.1.5.2 Critical analysis of the research item

The paper's strength lies in its integration of visual sensing and obstacle manipulation in the NAMO problem, which traditionally focused only on obstacle avoidance. By using

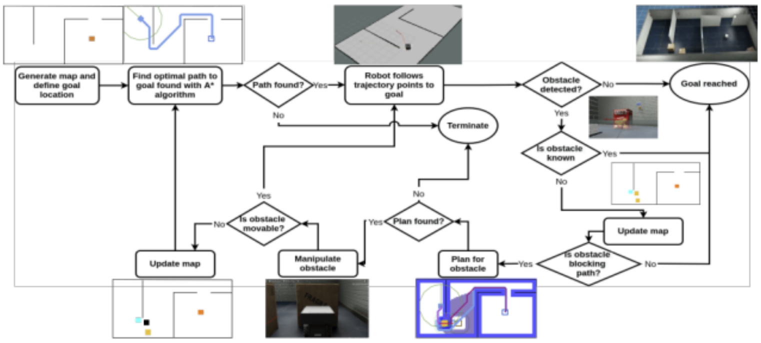

Fig. 2: Our developed Navigation Among Movable Obstacles (NAMO) pipeline.

Figure 2.4: NAMO PIPELINE

photorealistic simulation for training, the authors address the sim2real problem effectively, bridging the gap between simulation and real-world performance. However, one limitation is the focus on pushing as the only interaction, which may not be sufficient for environments with varied obstacles requiring different manipulation techniques. Additionally, the system's reliance on pre-trained object recognition models, such as DOPE, may limit its generalization to new environments with unseen objects.

2.1.5.3 Relationship to the proposed research work

This work is highly relevant to our research on robotic navigation and affordance learning in dynamic environments, especially in scenarios where robots need to interact with obstacles rather than just avoid them. The NAMO (Navigation Among Movable Obstacles) framework offers valuable insights for search and rescue operations, where robots must manipulate debris or objects to reach victims.

This aligns with our goal of using affordance concepts to enable robots to dynamically interact with their environment, enhancing their efficiency in complex and changing conditions. Additionally, NAMO's use of sim2real methods, which transfer simulation-based learning to real-world robotic systems, is crucial to our work. By using simulators like Gazebo for affordance learning and path planning, we can ensure that robots trained in simulation can perform effectively in real-world operations. This helps reduce the gap between simulation and actual performance, making our robots more reliable in disaster response missions.

In summary, the NAMO framework provides essential strategies for improving robotic interaction, path re-planning, and the transfer of learning from simulation to reality, all of which are vital for advancing our multi-agent robotic system.

Fig. 2: NUHome and the global map. The HSR is commanded to navigate from the green star to the orange star. The chair and the water bottle are not registered on the global map and the HSR is not aware of their existence beforehand.

Figure 2.5: NAMO

2.1.6 Affordance-Based Mobile Robot Navigation Among Movable Obstacles.[6]

2.1.6.1 Summary of the research item

This paper introduces a novel framework for autonomous mobile robot navigation among movable obstacles (NAMO). Traditional approaches to robot navigation focus on obstacle avoidance, which often leads to inefficient movements, particularly in cluttered and dynamic environments like homes. The proposed framework draws inspiration from human navigation behavior, where obstacles are not always avoided but sometimes moved. This is achieved through affordance-based navigation, allowing the robot to assess whether obstacles are movable and interact with them when necessary. The methodology combines affordance extraction and contact-implicit trajectory optimization (CITO) to enable robots to detect the movability of obstacles (pushability and liftability) and plan optimized paths that involve interacting with these obstacles. The framework is validated through experiments with Toyota's Human Support Robot in real-world scenarios, demonstrating its efficacy in environments where traditional obstacle avoidance would fail.

2.1.6.2 Critical analysis of the research item

The primary strength of the paper is its novel application of contact-implicit trajectory optimization and affordance-based reasoning, which enables better navigational similarity to humans. The combination of liftability and pushability affordances offers flexibility while maneuvering through crowded areas. Robust methodology that combines real-time interaction with affordance detection greatly enhances the robot's ability to navigate intricate settings. One drawback is the computational expense, which can impede real-time applications in more dynamic environments, particularly during affordance recognition and trajectory optimization. Furthermore, although the framework performs well in semi-dynamic and static contexts, its effectiveness in highly dynamic environments—that is, environments with constantly shifting obstacles—has not completely investigated. To improve its application, more computational process optimization is required, especially for the CITO.

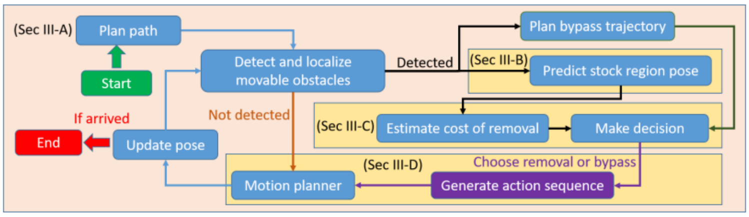

Figure 2.6: Namo optimaziton

2.1.6.3 Relationship to the proposed research work

This research paper is helpful in our works in this way that it use the concepts of NAMO, that is basically the affordance. we need that our robots interact with the object, to determine whether they are push able or not.After it can update the path that is used by this other robots or rescuers to save victim. This papers uses Namo and A* algorithm which is very useful us to decide a road map for over project.

2.1.7 Navigation among Movable Obstacles Using Machine Learning Based Total Time Cost Optimization.[7]

2.1.7.1 Summary of the research item

In this research paper author uses the same concepts of NAMO with the reinforcement learning technique.There are two main aspects of this paper ,first is cost optimization and second, making the path accessible for future goals.Firstly, the robot will determine the mobility of the obstacles.In this research robot will not just push the object aside, instead it will determine the total time cost of robot ,to determine whether bypassing or moving object aside will cost less.If moving the object aside makes the total time cost less,then it will check that placing a object aside doesn't create hurdle for future goals

2.1.7.2 Critical analysis of the research item

This research's primary strength is its creative application of machine learning to forecast the overall time cost and enhance decision-making in the NAMO assignment. The article offers a distinct edge above conventional approaches since it takes time efficiency into account instead of just avoiding or removing obstacles. Nevertheless, the method might encounter difficulties in extremely dynamic settings where the learnt model might find it difficult to adjust. Furthermore, the usefulness of the method may be limited in real-world scenarios where complexity is added by sensor errors and dynamic environments due to its dependence on simulations. Even while it shows promise, the reinforcement learning-based method may be computationally costly, particularly when scaling to larger environments or more agents.

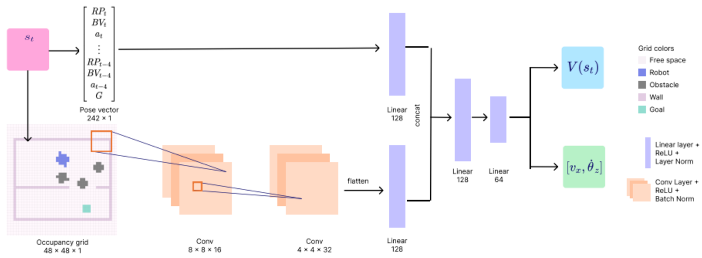

Figure 2.7: local path planning

2.1.7.3 Relationship to the proposed research work

The objective of optimizing robot performance in dynamic contexts, such those encountered in search and rescue missions, is directly supported by this research, which provides insightful information on the use of machine learning for time-efficient navigation. This study will help us enhance the decision-making capabilities of our rescue robots, which will help them respond to changing circumstances more skillfully a critical ability in high scenarios involving the risk of human life.
In conclusion, our research offers fundamental methods and strategies that can greatly improve our rescue robots' performance and flexibility, enabling them to traverse dynamic settings more successfully.

2.1.8 Local Path Planning Among Pushable Objects Based on Reinforcement Learning.[8]

2.1.8.1 Summary of the research item

In this research paper, they train multiple agents in a simutaled environ- ment using the A2C algorithm combined with the CNN. In training domain randomization concept is used to ensure that agents can handle unknown environment.The major set back of this ,is that it cannot handle real world complex problem.

2.1.8.2 Critical analysis of the research item

The primary strength of this research is its use of deep reinforcement learning to handle dynamic and uncertain environments, particularly through non-axis-aligned pushing. The method is robust in dealing with sensor noise and adapting to unseen obstacles, which is critical for real-world applications. However, a limitation is the potential difficulty in scaling this method to larger environments or more complex tasks, as training reinforcement learning agents can be computationally expensive. Furthermore, the use of simulations for training may lead to discrepancies when transitioning to real-world applications, despite the use of domain randomization to mitigate this gap. The paper also focuses heavily on pushing, while other forms of object manipulation like lifting or pulling are not considered.

2.1.8.3 Relationship to the proposed research work

This research directly contributes to the goal of enhancing local path planning in dynamic environments through the application of reinforcement learning. This aligns with our aim to develop more effective navigation strategies for our rescue robots operating in unpredictable scenarios. The process of teaching agents to push through barriers and adjust to shifting circumstances can help our proposed research produce more effective path-planning algorithms. By utilizing these strategies, our robots may be able to navigate through congested surroundings more adeptly by dynamically changing their routes in response to obstructions. In conclusion, our research provides important new understandings and techniques that can greatly improve our rescue robots' local path planning abilities and help them perform better in dynamic and unpredictable contexts.

2.2 Analysis Summary of research items

Here is the tabular representation for critical analysis of the research item discuss above

Research Item	Strengths	Weaknesses
Cerberus Transformer	Unique multitasking strategy for indoor scene interpretation, effective long-range dependency handling.	Heavily reliant on balanced task weights and demands significant processing power for training.
One-shot Open Affordance Learning with Foundation Models	Efficient limited data usage and generalization of novel affordances.	May struggle with complex affordances requiring deeper reasoning; potential bias from reliance on foundation models.
AffordanceNet	High efficiency and precision in real-time affordance identification, simplifies training and testing.	Limited generalization due to reliance on specified affordance classes; requires substantial processing resources.
APEX: Affordance-Based Plan Executor for Indoor Robotic Navigation	Affordance-based navigation enables adaptability in dynamic or unfamiliar environments without needing a detailed map.	Decreased performance in unseen environments and challenges in learning high-level behaviors, potentially introducing errors.
Navigation Among Movable Obstacles	Integrates visual sensing and obstacle manipulation, effectively addresses sim2real challenges.	Focuses solely on pushing interactions and relies on pre-trained models, limiting generalization.
Affordance-Based Mobile Robot Navigation	Novel trajectory optimization and affordance-based reasoning, enhances human-like navigation.	High computational expense impacts real-time performance; effectiveness in highly dynamic environments remains untested.
Navigation Among Movable Obstacles Using Machine Learning Based Total Time Cost Optimization	Creative application of machine learning for time cost optimization in obstacle navigation, improving decision-making.	May struggle in dynamic environments; real-world performance limited by sensor errors and simulation dependency.
Local Path Planning Among Pushable Objects Based on Reinforcement Learning	Robust in handling dynamic environments, non-axis-aligned pushing, and sensor noise, critical for real-world applications.	Potential scalability challenges; relies on simulations, leading to gaps in real-world transition despite domain randomization.

Table 2.1: Critical Analysis of Research Items

Chapter 3

Proposed Approach

Here is the template of architecture which we are going to use so far.

3.1 Overall Architecture

Architecture is designed to tackle three interrelated tasks: semantic parsing, affordance, and attribute parsing. The idea is that understanding these tasks together can improve performance, as objects in images often have related properties. n be moved, which is important for both understanding its semantic role and its affordance.

3.1.1 Transformer Encoder

- **Image Input:** An image of dimensions H×W (height and width) is taken as input.
- **Patch Division:** The image is divided into smaller, non-overlapping square patches, each of size p×p.
- **Token Formation:** Each patch is flattened and passed through a ResNet-50 backbone, generating embeddings.
- **Positional Information:** To keep track of where each token came from, learnable position embeddings are added to the tokens.

3.1.2 Reassemble Operation

After processing the tokens through the transformer:

- **Feature Representation:** The tokens are transformed into image-like feature representations. The embeddings are projected into a higher-dimensional space using a fully connected layer.
- **Rearranging**: The newly created features are rearranged based on their original positions in the image, forming a feature map.
- **Upsampling:** This feature map is then resized using a spatial unsampling layer to form a larger feature map Fupsample.

3.1.3 Fusion Block

The model generates feature maps at different resolutions from various stages (ResNet-50 blocks and transformer layers): **Feature Fusion:** Cerberus employs RefineNet-style fusion blocks to combine these feature maps. In each fusion stage:

- A Residual Convolutional Unit (RCU) processes the previous reassembled feature.
- This output is combined with features from the previous stage via element-wise summation and processed through another RCU.
- The combined feature map is then upsampled for the next stage.

3.1.4 Prediction Head

The final step involves generating predictions for each task:

- **Separate Heads:** There are three distinct heads for predicting semantic maps, affordance, and attributes.
- **Each head includes:**

1. A fully connected layer that produces the respective prediction map.

1. An interpolation function that upsamples the predicted map to match the original image's resolution.

- **Output Sizes:**
- For affordance and attributes, the output maps are y×H×W and z×H×W respectively (where y and z are the number of classes).
- The semantic map corresponds to H×W, where each pixel denotes a semantic class.
- **Loss Calculation:** The model employs binary cross-entropy losses for affordance and attribute predictions and a multi-class cross-entropy loss for semantic predictions to optimize the learning.

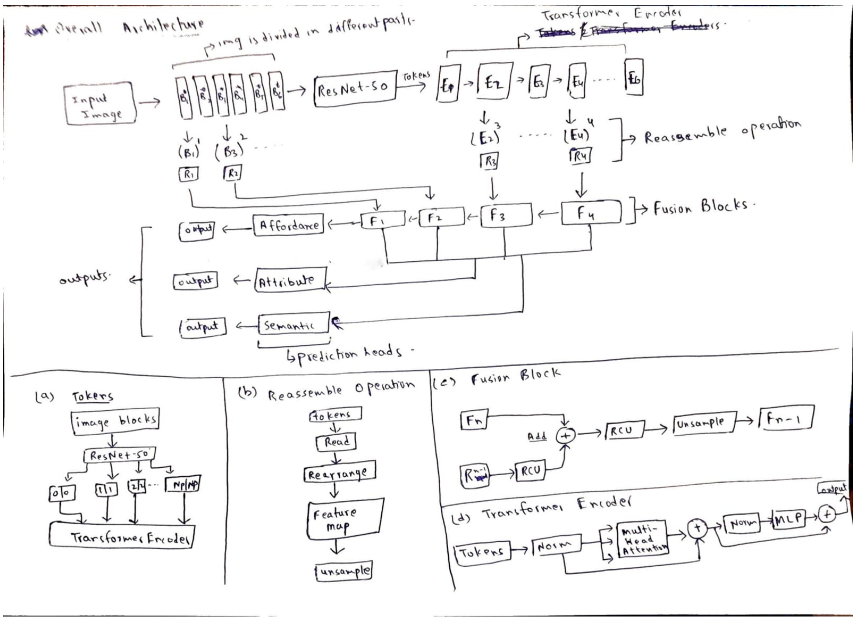

Figure 3.1: Architecture of affordances

Chapter 4

Conclusions and Future Work

Upon careful review and analysis of the research presented, we developed an architecture that effectively leverages the transformer model to process visual inputs. By capturing long-range dependencies and enabling shared representations across multiple tasks, the proposed design significantly enhances performance while maintaining flexibility for various dense prediction applications. Additionally, the integration of PATH planning principles provides a clear and optimized route, guiding rescuers more efficiently during critical operations.

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